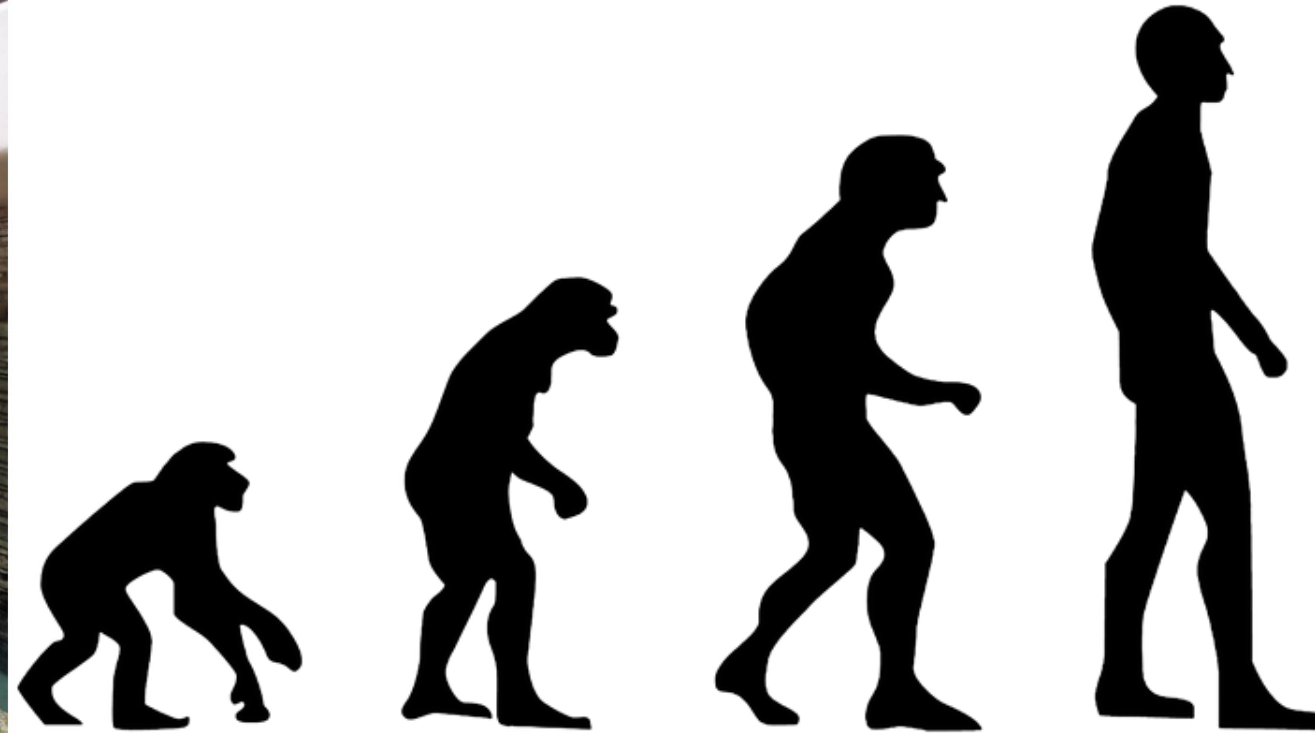


Basics and Principles

Outline

- Focus on Humans
- Mental Model, Conceptual Models, and Metaphors
- Human Centered Design
- Know the Users and Tasks



Focus on the Human

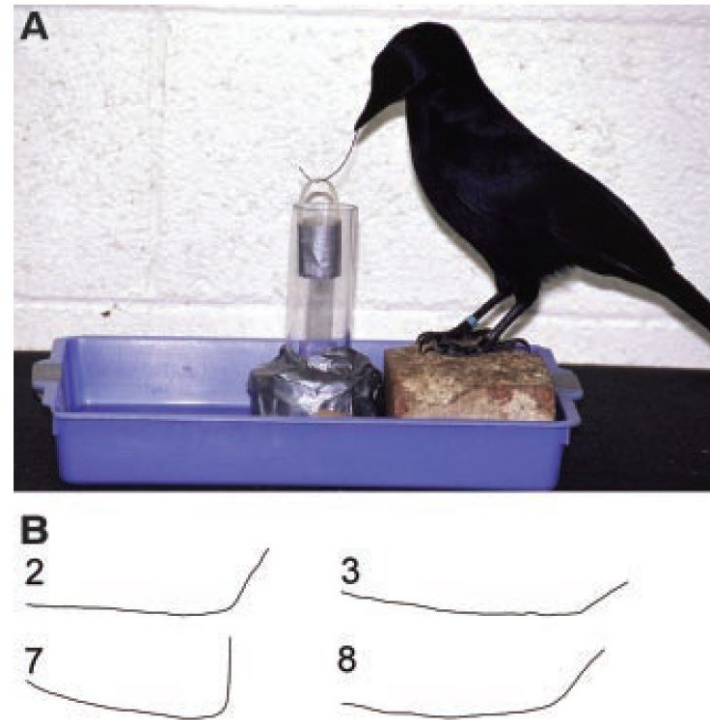
Learning Goals

- Understand ...
 - the basic activities in a human centered design process
 - what are potential products in the design process
 - what typical design objectives are
- Be able to
 - explain design products and design objectives in the context of a real world scenario
 - explain the key principles for usable systems by Gould and Lewis
 - discuss the statement “Make Mistake Early and Recognize Them” in the context of usability

We build tools that extend our abilities

Tool use, intelligence, and evolution

The way humans make and use **tools** is perhaps what **sets our species apart** more than anything else. Now scientists are more and more uncovering the forces that drove our lineage to our heights of tool use — and how **tool use**, in turn, might have **influenced our evolution.**”

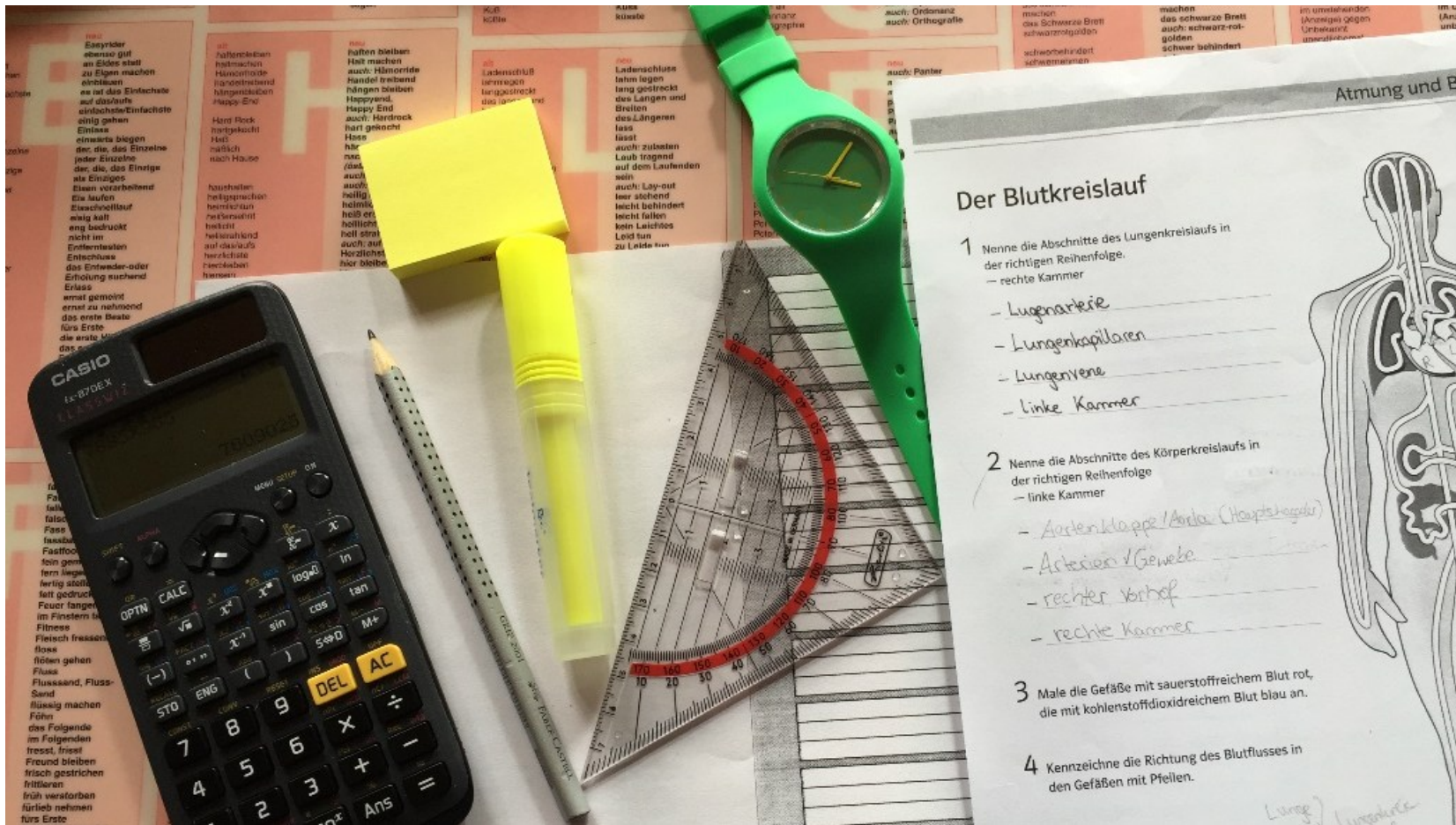


Charles Q. Choi. 2009. Human Evolution: The Origin of Tool Use Live Science, November 11, 2009.

Weir, A.A., Chappell, J. and Kacelnik, A., 2002. Shaping of hooks in New Caledonian crows. *Science*, 297(5583), pp.981-981.

Tools to Augment Cognition

Tool use, intelligence, and evolution



A collection of tools designed to support and enhance cognitive processes:

Example: learning and problem-solving related to biology (specifically, the circulatory system).

It is not Simple to Make Good Interactive Products!

Examples are all around us..., some basic misconceptions

- If I (the developer) can use it, everyone can use it
- If our non-technical staff can use it, everyone can use it
- Good designs/user interfaces are applied common sense
- A system is usable if all norms and style guidelines are met

That is why a process is required.

How it does NOT work

.... Don't fall for this

- Usability tests at the end when the product is ready and needs to be shipped
- Designing a new and pretty skin or look & feel to a product
- Introducing HCI issues after the system architecture and the foundations are completed
- Have the developers and testers test the usability of the system

Creating Interactive Systems

Structured process along the development cycle

- An interior designer can not make a great house if the architect and engineers forgot windows, set the doors at the wrong locations, and created an unsuitable room layout.
- Creating the user interface at the end of the project will not work – it has been considered from the very beginning.

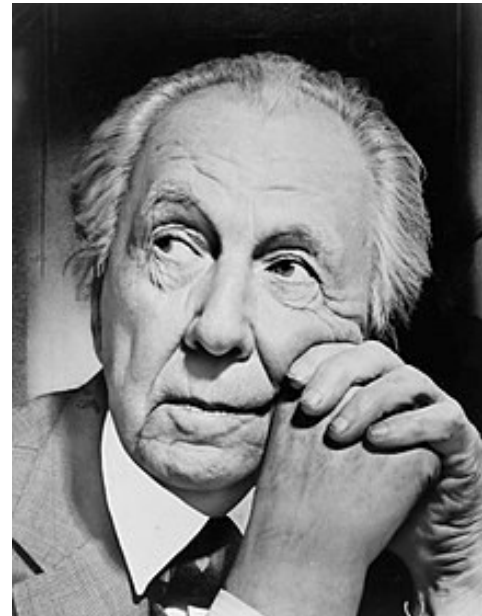


Make Mistake Early and Recognize Them!

Early involvement is key

Frank Lloyd Wright

“You can use an eraser on the drafting table or a sledgehammer on the construction site.”



Slide Adapted from Lewis Chuang

Focus on **Technology** or **User**?

Cylinder record or Disk record?

Both were designed to
record and reproduce sound



- **Graphophone:** Thomas Alva Edison (1880)



- **Gramophone:** Emile Berliner (1887)

Slide Adapted from Lewis Chuang

Focus on Technology or User?

Cylinder or Disk?

Benefits of the disc

- mass-production
- less storage space
- easier shipping
- allow double-sided recordings
- better marketing (famous artistes)



- Emile Berliner (1887)

Designing products

Key aspects of design

- **Interfaces:** The touchpoints between users and products
- **Interactions:** How users engage with a product
- **Environments:** The physical or digital spaces where a product is used
- **tasks and processes:** The steps users take to accomplish goals
- **user experience:** The overall feel of using a product
- **organizational structures:** How teams, roles, and workflows within a company influence design. Good structures enable collaboration and innovation.
- **Society:** The broader impact of a product, including ethical considerations, inclusivity, and cultural relevance.
- **Needs:** Addressing user pain points and desires



Why to design?

- create new products, systems, & experiences
- improve existing products:
- ensure safety: features and training
 - To protect users from harm through built-in safety mechanisms and proper education
- develop performance support
 - To assist users in real-time with tools, automation, and guidance
- develop methods for training and assessment
 - To help people acquire new skills effectively and measure progress
- guide team and organization formation
 - To structure teams for collaboration, efficiency, and innovation



Designing for Usability

Key Principles by Gould and Lewis, 1985

- “Any system designed for people to use should be **easy to learn** (and remember], **useful**, that is, contain functions **people really need** in their work, and be **easy and pleasant to use**.”
- “...three principles of system design which we believe must be followed to produce a useful and easy to use computer system [...]
 - Early Focus on Users and Tasks
 - Empirical Measurement : collecting and analyzing real user data
 - Iterative Design”

RESEARCH CONTRIBUTIONS

Human Aspects
of Computing

Henry Ledgard
Editor

Designing for Usability: Key Principles and What Designers Think

JOHN D. GOULD and CLAYTON LEWIS

ABSTRACT: This article is both theoretical and empirical. Theoretically, it describes three principles of system design which we believe must be followed to produce a useful and easy to use computer system. These principles are: early and continual focus on users; empirical measurement of usage; and iterative design whereby the system (simulated, prototype, and real) is modified, tested, modified again, tested again, and the cycle is repeated again and again. This approach is contrasted to other principled design approaches, for example, get it right the first time, reliance on design guidelines. Empirically, the article presents data which show that our design principles are not always intuitive to designers; identifies the arguments which designers often offer for not using these principles—and answers them; and provides an example in which our principles have been used successfully.

Any system designed for people to use should be easy to learn (and remember), useful, that is, contain functions people really need in their work, and be easy and pleasant to use. This article is written for people who have the responsibility and/or interest in creating computer systems (or any other systems) with these characteristics. In the first section of this article we briefly mention three principles for system design which we believe can be used to attain these goals. Our principles may seem intuitive, but system designers do not generally recommend them, as results of surveys reported in Section 2 show. The recommendations of actual designers suggest that they may sometimes think they are doing what we recommend when in fact they are not. In Section 3 we contrast some of their responses with what we have in mind to provide a fuller and clearer description of our principles. In Section 4 we consider why designers might not actually be using our design

principles. In Section 5 we elaborate on the three principles, showing how they form the basis for a general methodology of design. In Section 6 we describe a successful example of using our recommended methodology in actual system design, IBM's Audio Distribution System (ADS), and the advantages that accrued as a result.

1. THE PRINCIPLES

We recommend three principles of design.

Early Focus on Users and Tasks

First, designers must understand who the users will be. This understanding is arrived at in part by directly studying their cognitive, behavioral, anthropometric, and attitudinal characteristics, and in part by studying the nature of the work expected to be accomplished.

Empirical Measurement

Second, early in the development process, intended users should actually use simulations and prototypes to carry out real work, and their performance and reactions should be observed, recorded, and analyzed.

Iterative Design

Third, when problems are found in user testing, as they will be, they must be fixed. This means design must be iterative: There must be a cycle of design, test and measure, and redesign, repeated as often as necessary.

2. WHAT SYSTEM DESIGNERS AND PROGRAMMERS ACTUALLY SAY

We began recommending these principles in the 1970's. Often the reaction is that they are obvious. Nevertheless, they are not usually employed in system design. Why? We wondered whether or not these principles were really obvious, or whether or not they just

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March 1985 Volume 28 Number 3

Gould, J. D., & Lewis, C. (1985). Designing for usability: key principles and what designers think. *Communications of the ACM*, 28(3), 300-311.

Designing for Usability

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Iterative Design

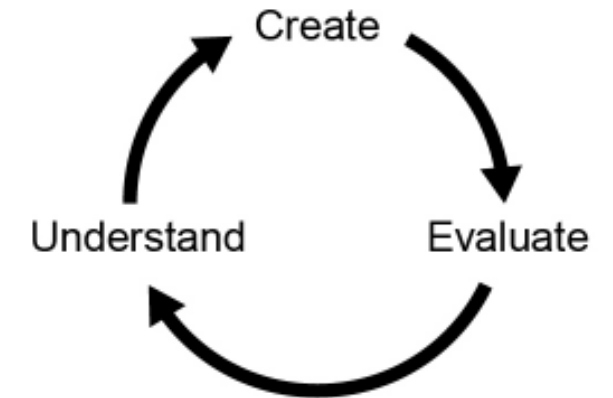
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Four basic activities of interaction design

- **Identifying needs** and establishing requirements for the user experience
- **Developing alternative designs** that meet those requirements
- **Building interactive versions** of the designs
- **Evaluating** what is being built throughout the process and the user experience it offers





Did you understand this lesson?

Can you answer these questions?

- Name the key principles for usable systems according to Gould and Lewis
- Give an example of an interactive systems design, that shows the importance of the statement “Make Mistake Early and Recognize Them”
- Describe typical aspects of product design and design objectives.
- What are the basic activities in a human centered design process?

References

- Pavel, A. (1983). U.S. Patent No. 4,412,106. Washington, DC: U.S. Patent and Trademark Office.
<http://www.google.com/patents/US4412106>
- Gould, J. D., & Lewis, C. (1985). Designing for usability: key principles and what designers think. Communications of the ACM, 28(3), 300-311.

Mental Modal and Metaphors

Learning Goals

- Understand ...
 - Why it is hard for developers to spot usability issues.
 - How mental model, conceptual model, and metaphor relate
- Be able to explain
 - the terms mental and conceptual model,
 - The difference between mental and conceptual model,
 - what metaphors are, why they are useful, and where their limitations are.

What is a Mental Model

How the user thinks it works

‘In human-computer interaction research, the notion of "**mental models**" has come to be a very general catch-phrase for **anything having to do with end users' knowledge** of an application (van der Veer, 1990).

Optimistic View:

- There is a feeling that **if we could "capture" mental models, then we could build good interfaces [...]**

Skeptical View:

- But many [...] are much less convinced of the alleged benefits of mental models and of our ability to use them for reasoning or other complex cognition. [...] Researchers who have investigated mental images have been struck by how **incomplete and inflexible they are...**

Nardi, B. A., & Zamer, C. L. (1993). Beyond models and metaphors: Visual formalisms in user interface design. *Journal of Visual Languages & Computing*, 4(1), 5-33.

Mental Model

How the user reasons and understands

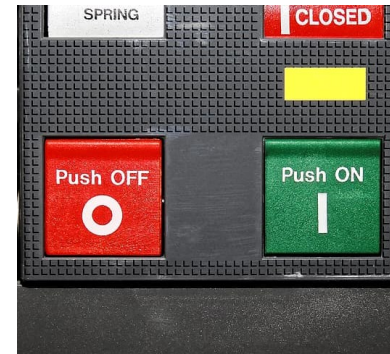
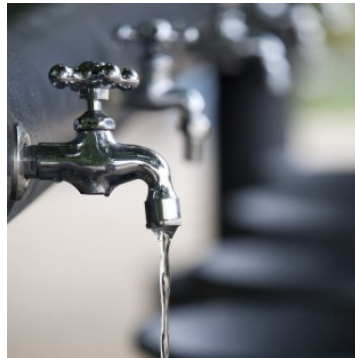
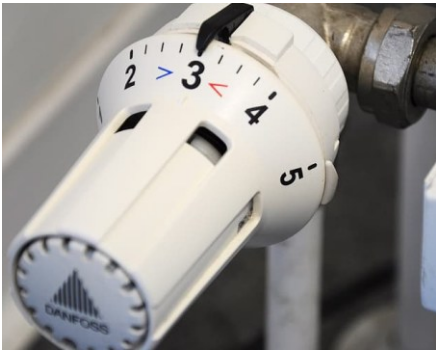
- Kenneth Craik (1943) 'the mind constructs "small-scale models" of reality that it uses to anticipate events, to reason, and to underlie explanation'
- Users acquire mental models by
 - Interaction / observation
 - Explanation
- Two types
 - Functional – users know what to do, but not why
 - Structural – users know why to do something

William Hudson. Mental Models, Metaphor and Design (2003).UK UPA and HCI2003
<http://www.syntagm.co.uk/design/articles/mmmad.pdf>

Mental Model

Example: Central Heating

- You came back and it is only 15°C in your room. You want a room temperature of 21°C. What do you do?
 1. You put the thermostat to 21°C and wait
 2. You put the thermostat to max (=35°) and switch it back to 21° once it is warm?
- What different mental models do people have who chose strategy 1 or 2?



William Hudson. Mental Models, Metaphor and Design (2003).UK UPA and HCI2003
<http://www.syntagm.co.uk/design/articles/mmmad.pdf>

Mental Model

Definition by Jakob Nielsen

'A mental model is **what the user believes** about the system at hand.'

- 'A mental model is based on **belief, not facts**: that is, it's a model of what users know (or think they know) about a system such as your website. Hopefully, users' thinking is closely related to reality because they **base their predictions** about the system on their mental models and thus plan their future **actions** based on how that model predicts the appropriate course.'
- 'It's a **prime goal for designers** to make the user interface **communicate the system's basic nature** well enough that users form reasonably accurate (and thus useful) mental models.'

Jakob Nielsen. Mental Models on October 17, 2010
<https://www.nngroup.com/articles/mental-models/>

Mental Model

Definition by Jakob Nielsen

- 'Individual users each have their own mental model. [...] different users might construct different mental models of the same user interface. Further, one of usability's big dilemmas is the common **gap between designers' and users' mental models**. Because designers know too much, they form wonderful mental models of their own creations..., leading them to believe that each feature is easy to understand. Users' mental models of the UI are likely to be somewhat more deficient, making it more likely for people to make mistakes and find the design much more difficult to use.

Jakob Nielsen. Mental Models on October 17, 2010
<https://www.nngroup.com/articles/mental-models/>

Conceptual Model

The model the designers wants the user to have

- “A conceptual model is a high-level description of how a system is organized and operates.”

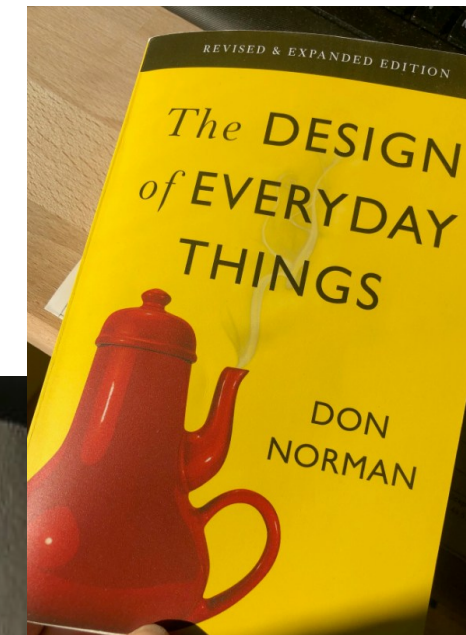
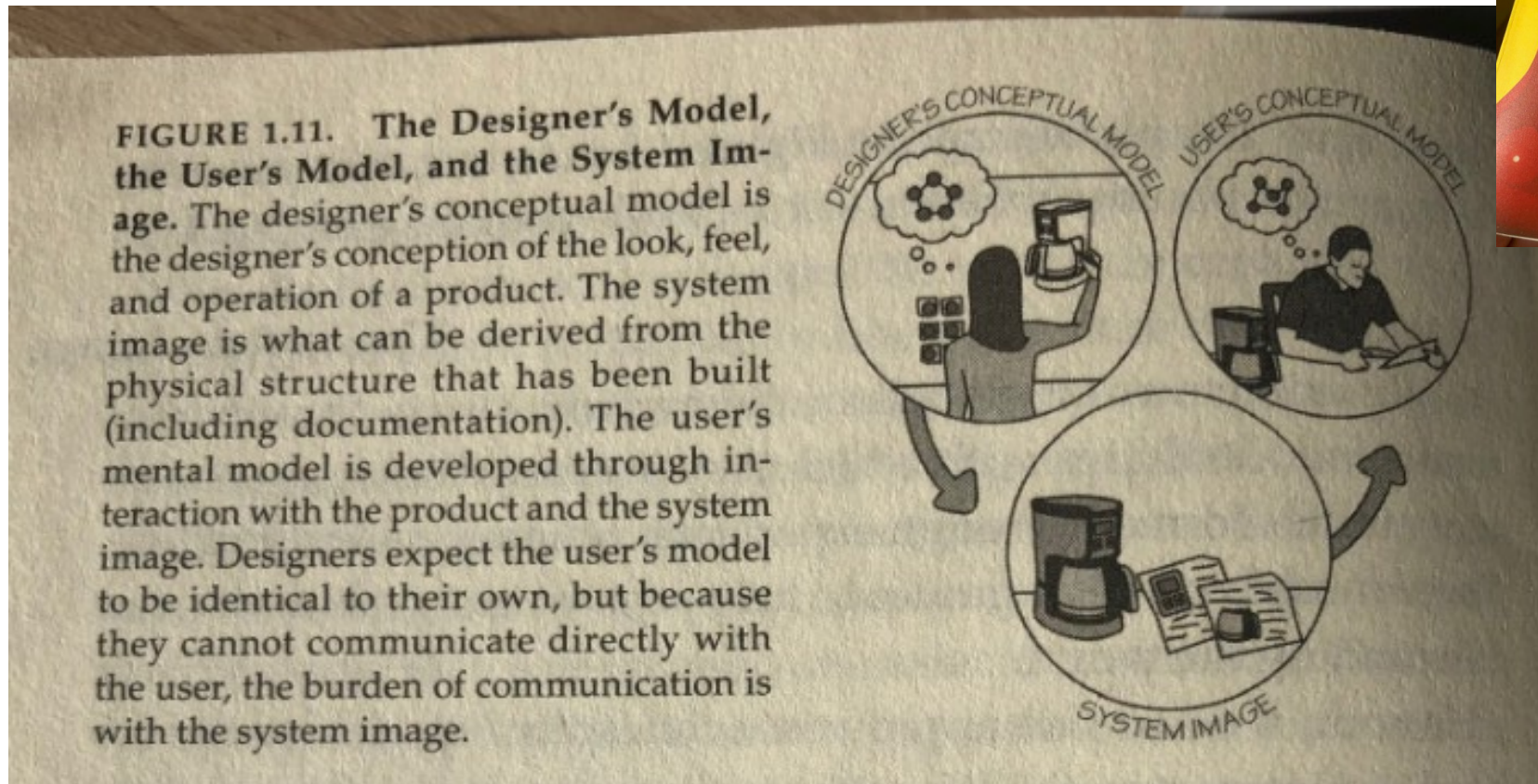
Johnson, J., & Henderson, A. (2002). Conceptual models: begin by designing what to design. *interactions*, 9(1), 25-32.

- The conceptual model
 - is deliberately designed
 - it allows to user to understand and operate the UI
 - it draws on prior knowledge of the user
 - is communicated through the interface and interaction design

William Hudson. Mental Models, Metaphor and Design (2003).UK UPA and HCI2003
<http://www.syntagm.co.uk/design/articles/mmmad.pdf>

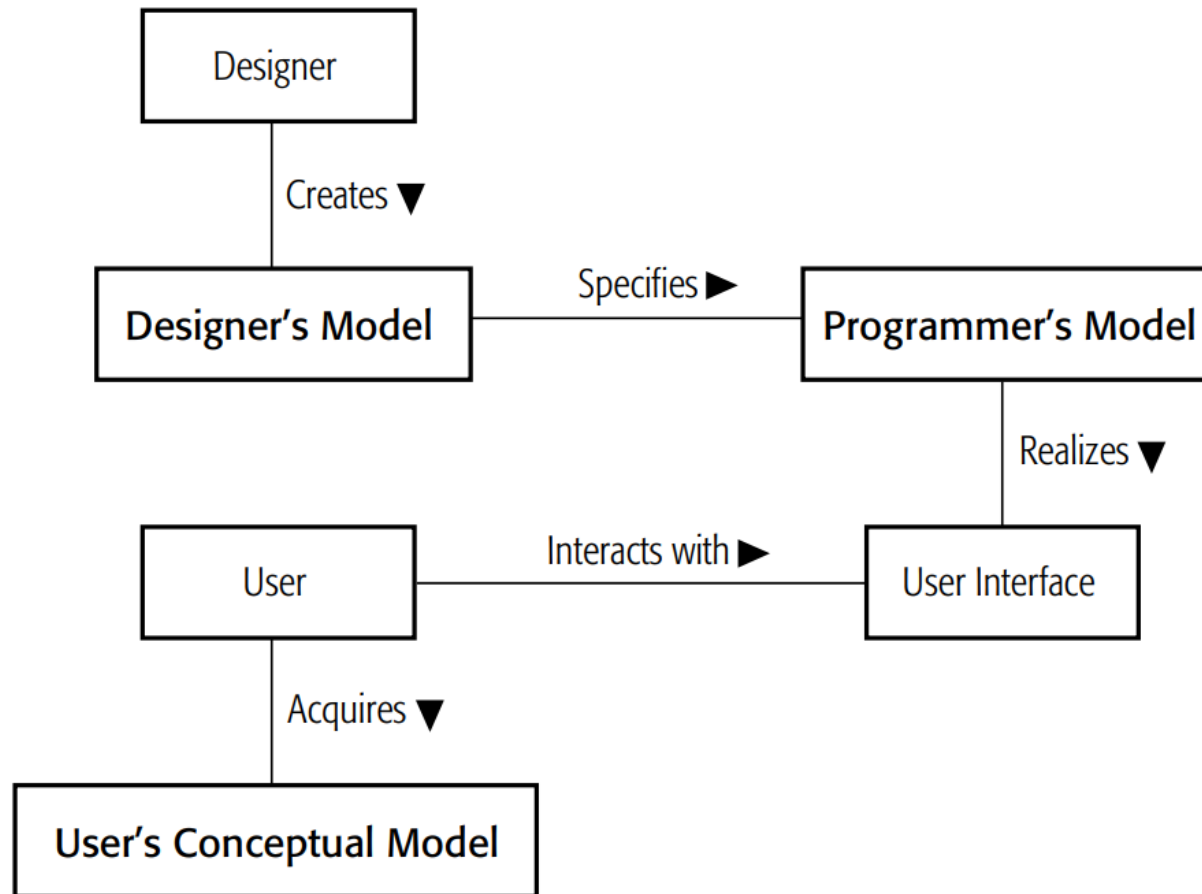
Designer Model, User Model, and System Model

Don Norman



Norman, D. A. (2013). The design of everyday things: Revised and expanded edition. New York: Doubleday.

Models – Designer, Programmer, User



Hudson, W. (2001). Toward unified models in user-centered and object-oriented design. Object Modeling and User Interface Design: Designing Interactive Systems, 313-362.

Clarification of Terms

Different things with the same intention

- **Mental Model** = User Model = User's Conceptual Model
- **Conceptual Model** = Designer Model
- **System Model** = Programmer's Conceptual Model = Programmer's Model = Implementation Model

Metaphors

Build on what the users know

‘Given the analogy, the new **user can draw upon his knowledge about the familiar situation** in order to reason about the workings of the mysterious new computer system. For example, if the new user wants to **understand about how the computer file system works**, he need only **think about how an office filing cabinet works** and then carry over this same way of thinking to the computer file system’

Frank Halasz and Thomas P. Moran. 1982. Analogy considered harmful. In Proceedings of the 1982 Conference on Human Factors in Computing Systems (CHI '82). Association for Computing Machinery, New York, NY, USA, 383–386. DOI:<https://doi.org/10.1145/800049.801816>

Metaphors

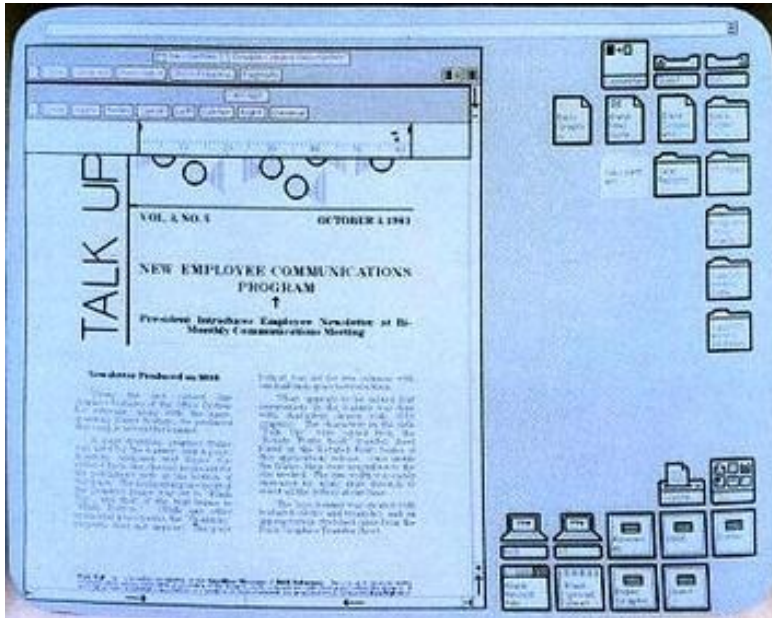
Build on what the users know

‘If people employ metaphors in learning about computing systems, the designers of those systems should anticipate and support likely **metaphorical constructions to increase the ease of learning and using** the system.’

Carroll, J. M., & Thomas, J. C. (1982). Metaphor and the cognitive representation of computing systems. IEEE Transactions on systems, man, and cybernetics, 12(2), 107-116.

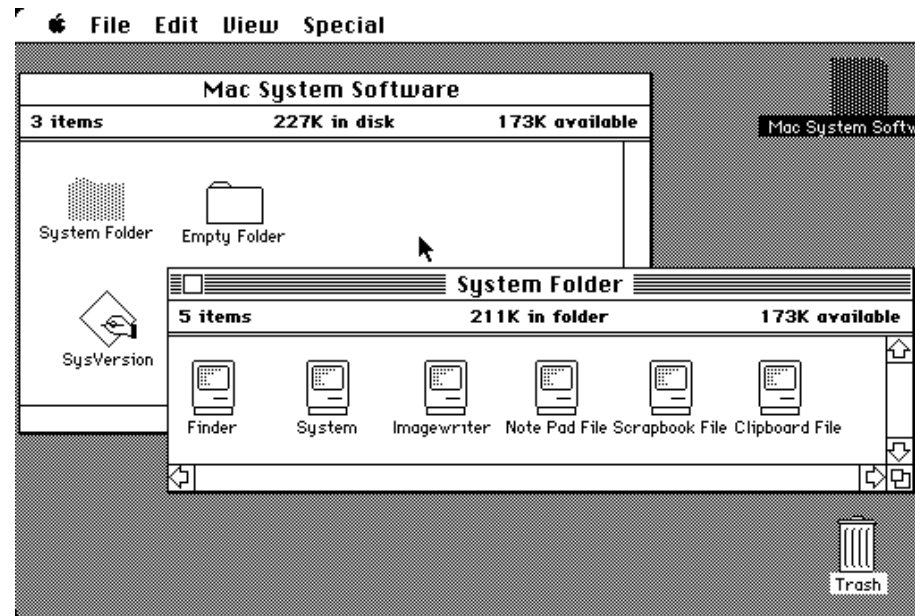
Metaphors - Examples

Desktop Metaphor – Alan Kay at Xerox Parc



Alan Kay's desktop metaphor is a user interface concept that revolutionized personal computing by making it more intuitive and accessible to non-technical users.

It's based on the idea of simulating a real-world desktop environment on a computer screen.





Did you understand this lesson?

Can you answer these questions?

- What is a mental model?
- What is the difference between functional and structural mental models?
- What is a conceptual model?
- What is the difference between a mental and a conceptual model?
- What are metaphors? What are their advantages and what are their limitations?

References

- Nardi, B. A., & Zamer, C. L. (1993). Beyond models and metaphors: Visual formalisms in user interface design. *Journal of Visual Languages & Computing*, 4(1), 5-33.
- William Hudson. *Mental Models, Metaphor and Design* (2003). UK UPA and HCI2003. <http://www.syntagm.co.uk/design/articles/mmmad.pdf>
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- Carroll, J. M., & Thomas, J. C. (1982). Metaphor and the cognitive representation of computing systems. *IEEE Transactions on systems, man, and cybernetics*, 12(2), 107-116.



Know the User and Tasks

Learning Goals

- Understand ...
 - Why it is important to know the user and their tasks
 - How a task frequency analysis works and what it is good for
 - The concept of personas and their value for designing interactive systems
- Be able to ...
 - explain why it is important to determine the supported task in the process before the system is designed
 - explain a hierarchical task analysis for a given example

Who is my user?

Usage Profiles: “Know Thy User”

- Different people have different requirements for their interaction with computers.
- Issues to take into account:
 - goals, motivation, personality
 - education, cultural background, training
 - age, gender, physical abilities, ...
- Experience:
 - Novice users
 - Knowledgeable intermittent users
 - Expert frequent users

Wilfred J. Hansen. 1971. User engineering principles for interactive systems. In Proceedings of the fall joint computer conference. <http://doi.acm.org/10.1145/1479064.1479159>

User-Needs and Task Profiles

What is the system for?

- Find out what the user is trying to do!
 - **the goal**
 - what their **needs** are
 - resulting **tasks**
- Supported tasks should be determined before the design starts
- Functionality should only be added if identified to help solving tasks
 - Temptation: If additional functionality is cheap to include it is often done – this can seriously compromise the user interface concept (and potentially the whole software system)!
- Frequency of tasks related to user profiles

User Diversity

One size fits all?

- Example: flight booking webpage
 - Travel agent booking many flights a day – everyday
 - A teacher organizing a field trip (once a year) and making bookings for a large group
 - A business person changing bookings while travelling
 - A family looking for a package holiday
- Basic concepts to structure the problem
 - **Usage profiles:** how different user groups interact with a product or service.
 - **Task profiles:** how a user performs a specific task within a given system or environment



Task Frequency Analysis

Hypothetical Analysis – travel web site

- Will one website fit all the users?

<i>Task</i>	Group reservation	Change of itinerary	Booking child care	Comparing sales agent performance
<i>Persona</i>				
Sales agent	0.2	0.1	0.1	0
Manager	0	0	0	0.3
Family	0.05	0.05	0.3	0
Business traveler	0.01	0.2	0.01	0

The frequency analysis helps prioritize which tasks are most important to support for each user group.

Task Frequency Analysis

Mini Exercise: Instagram, YouTube, EBay, WhatsApp

- Pick one Application, think of typical personas and their tasks, fill in hypothetical task frequencies

<i>Task</i>				
<i>Persona</i>				

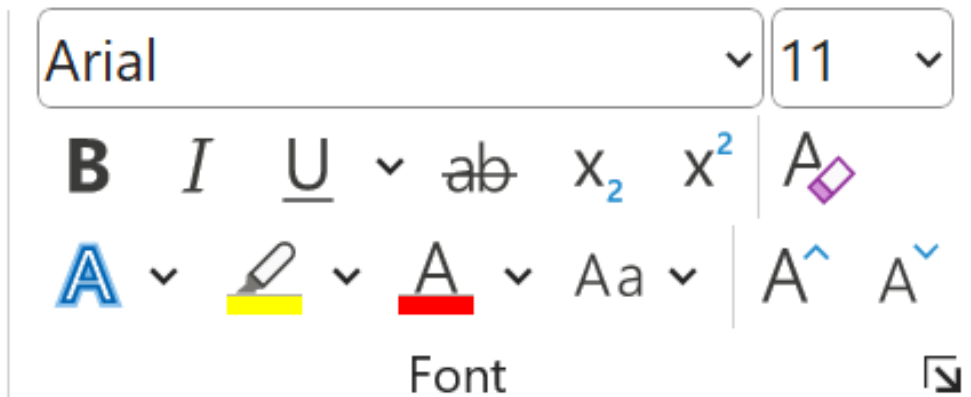
Task Frequency Analysis

What is it good for?

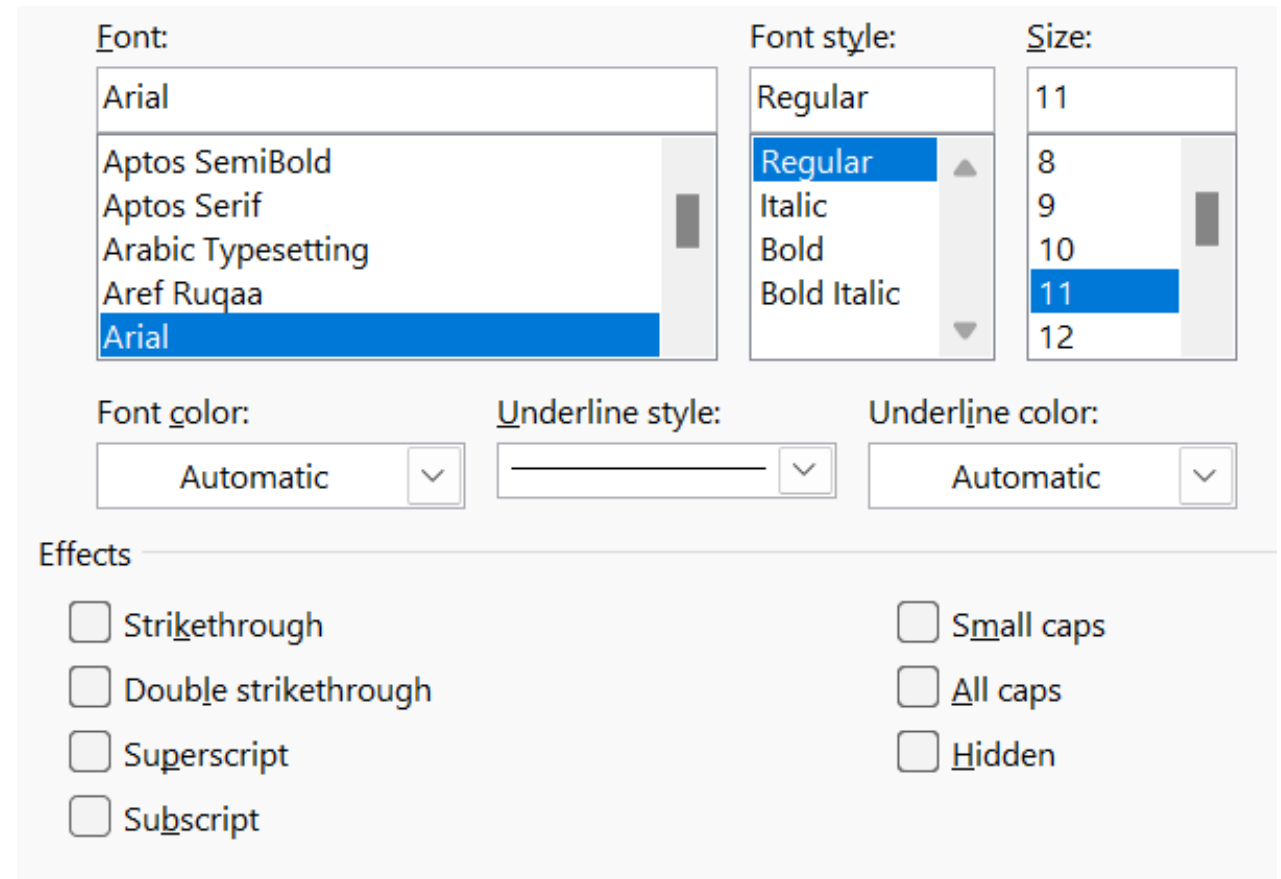
- Helps to shape the interaction design
 - Frequent action should be simple and quick to carry out
 - Infrequent action may take longer
 - Menu structure should reflect this
- Example
 - Frequent actions: Toolbar or special key
 - Intermediate frequent actions:
ribbon menu, pull-down menu, key combination (Ctrl+S)
 - Infrequent actions: Sequence of menus or dialogs
- Problem – if many (all) actions occur with very similar relative frequency...

Task Frequency Analysis

Example: **Bold** is more commonly used than **All Caps**

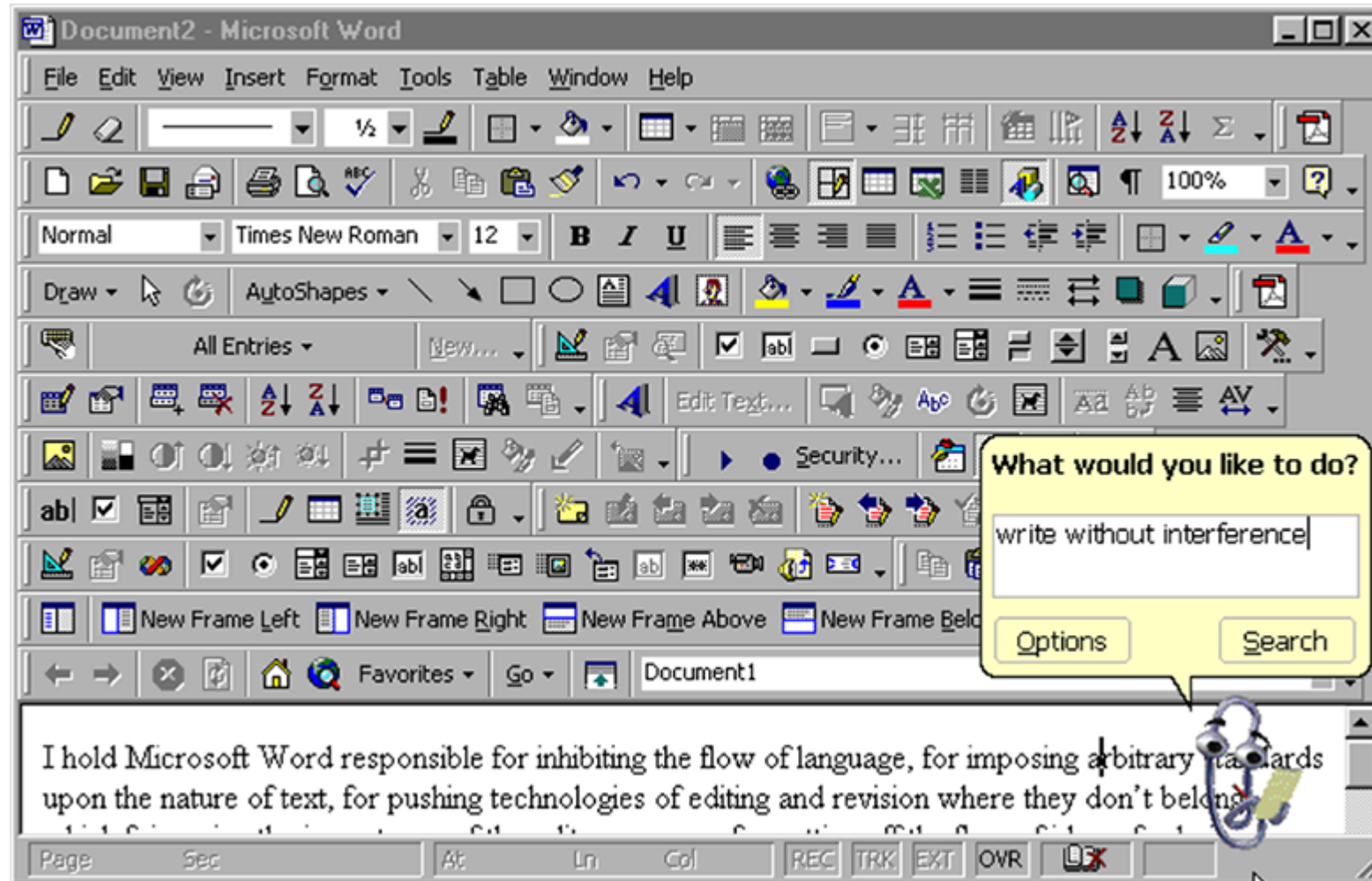


- More frequently used items in the toolbar
- Less frequently used items in Menus and dialog boxes



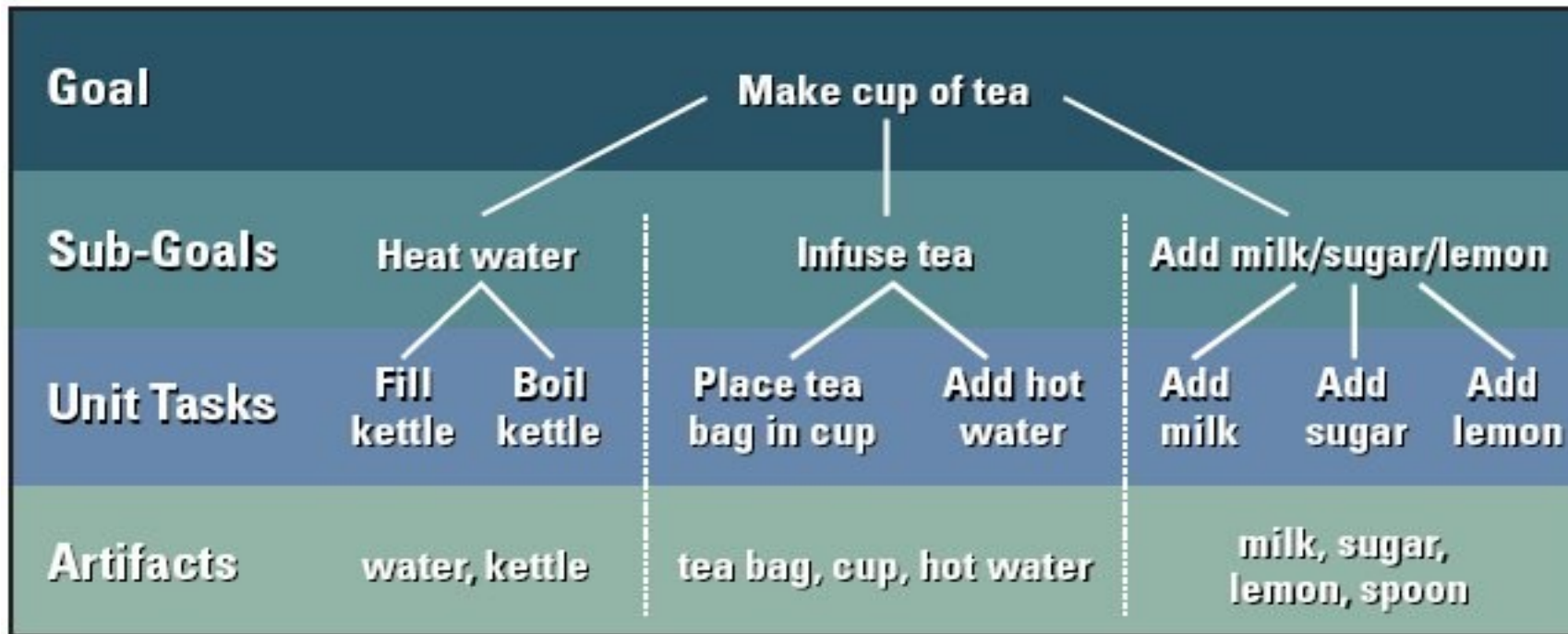
Task Frequency Analysis

Trade-off between quick access and over-crowded interface



Hierarchical Task Analysis

Understanding what it takes to complete a task



William Hudson. HCI and the Web: a Tale of Two Tutorials: a Cognitive Approach to Interactive System Design and Interaction Design Meets Agility. *ACM interactions* 12(1), 2005, 49-51

Where is the Average User?

- There is no such thing as an average user!
- Usage of Persona
 - Typical users but not the average
- Number game...
 - **Niche Targeting:** Creating a software that is appropriate for a very specific target group (e.g. 0,1% of the population) may still find a large user base, e.g. in Europe and the US this may be more than half a million people
 - **Mass Appeal:** Designing for 90% of the users will leave in Germany alone 8 million users out



Persona

Creating realistic and prototypical users

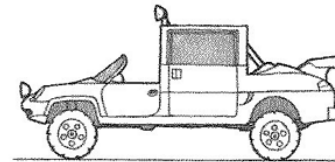
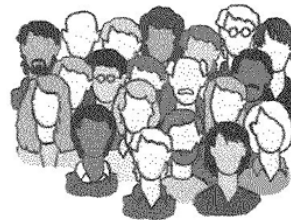
- “Software today is too often designed **to please too many users**, resulting in **low user satisfaction**.”
- “choose the right individuals or user groups to design for [...] and then to prioritize these individuals/groups so that the needs of the most important users are met”
- “Personas provide a powerful tool for communicating about **different types of users and their needs**, then deciding which users are the most important to target in the design of form and behavior”
- “Personas represent groups of users”
- Typically 3 to 12 personas

Cooper, A., Reimann, R., & Cronin, D. (2007). About face 3: the essentials of interaction design. John Wiley & Sons. p.80ff

Personas for Car Design

Example from A. Cooper et al.

- Personas and goals
 - Alesandro (fast and fun)
 - Marge (safe, comfortable)
 - Dale (big loads, reliable)
- Results in 3 designs that differ, e.g.
 - Porsche
 - City SUV
 - Pick-up
- Average
 - A car that is:
 - fast, fun, safe, comfortable, for big loads, and reliable
- The car that fulfils all requirements, will not fulfil them well...



Cooper, A., Reimann, R., & Cronin, D. (2007). About face 3: the essentials of interaction design. John Wiley & Sons. p.80ff



Did you understand this lesson?

Can you answer these questions?

- Why it is important to know the goals and tasks of the users?
- Alan Cooper argues “Software today is too often designed to please too many users, resulting in low user satisfaction.”. How can Persons in the design process help to address this problem?
- Conduct a task frequency analysis for a smart TV that is shared by a family
- Explain what personas are.
- Why is it not useful to design for an average user?
- Conduct a hierarchical task analysis for a taking a photo with your phone and posting it on Instagram.

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